BANGLADESH UNIVERSITY OF BUSINESS & TECHNOLOGY (BUBT)

Department of Computer Science and Engineering

Lab Experiment: 02

Software Development Approach Selection

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1. Objective

To study various software development models and select the most suitable model for the project titled **ZeroBuildStdio**. The objective is to evaluate the characteristics of available SDLC models against specific requirements, constraints, and the nature of the project.

2. Selected Model: Incremental Development Model

The **Incremental Development Model** was selected for the ZeroBuildStdio project. In this model, the system is built and delivered in a series of increments, where each increment adds new functionality to the previous working version. Rather than delivering the complete system at the end of development, each increment produces a testable and demonstrable application.

ZeroBuildStdio is divided into three key increments:

- **Increment 1 (Core Platform):** Auth & Admin module (login/register, admin dashboard, role management), HTTP server setup, and project configuration.
- **Increment 2 (Build & Manage):** Builder module (drag-and-drop theme customizer, product CRUD, base64 media pipeline), and shop configuration.
- **Increment 3 (Complete System & Deployment):** Storefront dynamic SSR rendering, checkout flow, Firebase persistence sync, and testing.

3. Comparison of SDLC Models

Factor	Waterfall	Spiral	Incremental	Agile
Requirement Flexibility	Low	High	Medium-High	Very High
Early Working Software	No	Partial	Yes	Yes
Risk Handling	Late	Strong	Incremental	Continuous
Process Overhead	Low	High	Medium	Medium
Fit for Fixed Deadline	Good	Poor	Good	Moderate
Fit for Student Team	Fair	Poor	Good	Good

4. Tool Used for Project Management

Jira Software & GitHub are used as the primary tools for tracking project increments, user stories, tasks, and commits for ZeroBuildStdio. Each increment is mapped to dedicated Jira Epics and GitHub pull requests.